

OSTEO OPTIX

- Problem Statement ID - SIH26004
- Problem Statement Title - AI-Assisted Early Detection System for Osteoarthritis Risk Markers in North Eastern Region (NER)
- Theme- Space Technology
- Category- Hardware
- Team ID -
- Team Name - RoboniX



RobonikX

SMART INDIA
HACKATHON
2026

Why current screening falls short

High latency

Radiographic Kellgren-Lawrence grading depends on slow, centralized imaging workflows.

Costly and centralized

Screening sits in imaging centers that remote and rural patients struggle to reach.

Blind to dynamic movement

A static image cannot capture ambulatory biomechanics: gait, lateral sway, weight bearing.

Prototype

Proposed Solution

Slow, centralized screening

Real-time scoring at the point of care

Static images miss movement

Dynamic ROM, gait sway and heel force

Rural, language and network gaps

Offline-first and Bhashini-localized

Three fused data pillars deliver one offline severity score

Hardware kinematics

ESP32 + BLE sleeve

Joint ROM, impact force and foot-strike cadence

40% of score

Edge AI vision

MediaPipe on-device

Gait asymmetry from 33 landmarks at 30 FPS

35% of score

Clinical context

WOMAC + Bhashini

Pain, stiffness and function in the local language

25% of score

Fusion engine → Total Severity Score (0-100)

Edge-compute

Offline-first

Multilingual

INNOVATION AND UNIQUENESS

Three-modality fusion

Sensors, vision and WOMAC, weighted

Zero-connectivity edge AI

Inference and scoring stay on the device

Detachable Brain Pod

Costly electronics apart from the washable sleeve

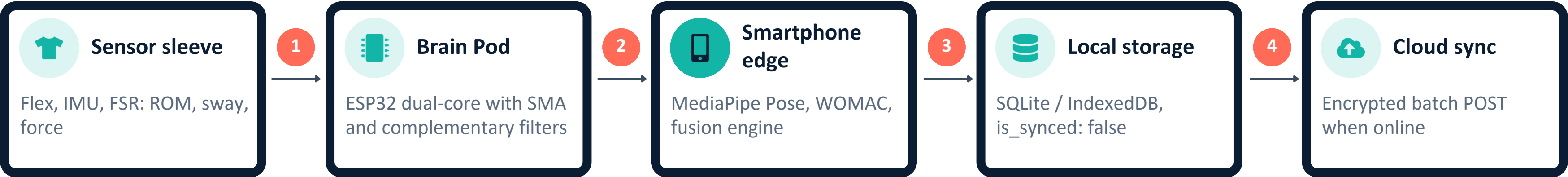
Localized survey

Regional-language WOMAC via Bhashini

prototype image

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1 10-pin magnetic pogo 2 BLE 4.2 GATT telemetry 3 JSON inference payload 4 Encrypted batch POST (online)

CORE ALGORITHM

$$Total\ Severity = (W_{gait} \times S_{gait}) + (W_{sens} \times S_{sensor}) + (W_{clin} \times S_{womac})$$

All inputs normalized to a 0-100 scale first

Worked example (illustrative)

Component	Score S	Weight W	W × S
Wearable sensors	70	0.40	28.0
Camera gait	60	0.35	21.0
WOMAC survey	50	0.25	12.5
Total severity			61.5

TECHNOLOGIES

Signal filtering
10-point SMA (Flex, FSR); complementary filter $\alpha = 0.98$ (IMU)

Knee angle
atan2 on hip, knee and ankle landmarks at 30 FPS

Symmetry Index
 $|Left - Right| / \text{mean of both} \times 100\%$

Severity fusion
 $0.40 \text{ sensors} + 0.35 \text{ gait} + 0.25 \text{ WOMAC}$

Connectivity

- BLE 4.2 GATT
- I2C at 400 kHz
- 12-bit ADC

Edge AI

- MediaPipe Pose
- LiteRT / TFLite
- Mobile GPU, 30 FPS

Data and language

- SQLite / IndexedDB
- ServiceWorker sync
- WOMAC, Bhashini REST

Hardware

- ESP32 LX6, FreeRTOS
- C++ scalar DSP
- 2S Li-Po, LM2596, LDO

Sensing

- Flex (carbon substrate)
- MPU6050 6-DOF IMU
- FSR402 force sensor

Clinical and language

- WOMAC 24-item index
- Bhashini NLP REST APIs
- Weighted severity fusion

DETAILED METHODOLOGY FOR EACH STAGE

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FEASIBILITY ANALYSIS



Technically feasible
Proven off-the-shelf parts: ESP32, MPU6050, flex and force sensors, MediaPipe.



Economically viable
Costly electronics sit in one reusable pod; only the low-cost sleeve is worn.

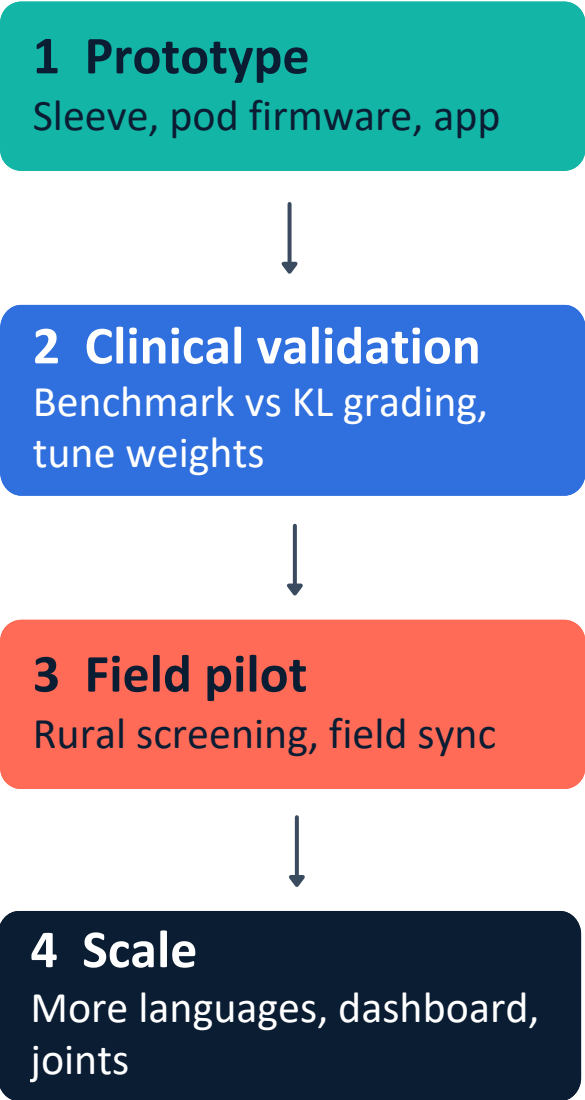


Operationally practical
Works at the point of care with zero connectivity, in the local language.

CHALLENGES AND STRATEGIES TO OVERCOME THEM

Challenge / risk	Strategy
Noisy sensor data, false positives	On-device SMA and complementary filters before BLE
Unreliable connectivity	Offline-first storage with queued background sync
Language barriers	Bhashini-localized WOMAC with cached translations
Patient data privacy	On-device inference, encrypted payloads, no frames sent to cloud
Single-modality error	Three fused modalities, weighted toward physical telemetry

ROADMAP





Zero-connectivity inference
Scoring runs on the device with no cloud offload.



Privacy by design
Camera frames stay on the phone; payloads travel encrypted.



No data loss
Queued records wait out volatile Tier-3 networks.

ONE SENSING ARCHITECTURE → MULTIPLE MUSCULOSKELETAL APPLICATIONS

IMPACT ON THE TARGET AUDIENCE

**Patients**

- Screening close to home
- Survey in their own language
- No travel to central imaging

**Clinicians and health workers**

- Objective ROM, gait and force data
- One 0-100 severity score
- Faster, evidence-based triage

**Health system**

- Decentralized diagnostics
- Records sync when online
- Scalable across regions

BENEFITS OF THE SOLUTION

**Social**

- Early detection in rural and remote areas
- Local-language access to screening
- Narrows the urban-rural care gap

**Economic**

- Reusable pod spreads electronics cost
- Low-cost washable sleeve
- Less reliance on central imaging

**Environmental**

- Modular pod avoids disposable devices
- Washable, reusable sleeve
- Fewer long trips to central facilities

Aligned with

SDG 3 | Good Health and Well-being

SDG 9 | Industry, Innovation and Infrastructure

SDG 10 | Reduced Inequalities

Digital India | Bhashini language mission

A screening system that travels to the user – not only a user travelling to the hospital

Reference and link	Supports
Kellgren JH, Lawrence JS. Radiological assessment of osteo-arthrosis. Annals of the Rheumatic Diseases, 1957;16(4):494-502.	Radiographic KL grading: the standard we position against
Bellamy N, et al. Validation study of WOMAC. Journal of Rheumatology, 1988;15(12):1833-1840.	24-item WOMAC index and 0-96 scoring
Bazarevsky V, et al. BlazePose: On-device Real-time Body Pose Tracking. arXiv:2006.10204, 2020.	33-landmark on-device pose tracking
Google AI Edge: MediaPipe Pose Landmarker and LiteRT. ai.google.dev/edge	Pose runtime on the phone GPU
Bhashini, National Language Translation Mission, MeitY, Govt. of India.	Regional-language localization of WOMAC
Espressif ESP32 Series Datasheet; InvenSense MPU-6000/6050 Product Specification.	Dual-core MCU, ADC, I2C and IMU ranges
Osteo Optix Engineering Specification v2.1	Architecture, DSP filters and fusion weights

Capability	Conventional (KL radiography)	Osteo Optix
Where it runs	Centralized imaging facility	On the device, at the point of care
Movement	Static image of the joint	Dynamic gait, ROM and weight bearing
Data	Single modality (radiograph)	Three fused: sensors, vision, WOMAC
Cost and speed	Expensive, high latency	Low-cost hardware, real-time results
Connectivity	Relies on facility infrastructure	Zero-connectivity; syncs later
Language access	Not built in	Regional languages via Bhashini



- **WOMAC INDEX** : <https://www.physio-pedia.com>
- **MEDIAPIPE DOCUMENTATION** : <https://research.google/pubs/mediapipe-a-framework-for-perceiving-and-processing-reality/>
- **OSTEOARTHRITIS** : <https://www.ncbi.nlm.nih.gov/>
- **DATA ANALYSIS** : <https://necouncil.gov.in/>
- **AI/ML VALIDATION , GAIT BIOMECHANICS , PLANTAR PRESSURE** etc.